

Chapter Overview**17.1 Visual Data Manager**

17.1.1 Creating Database in Visual Data Manager

17.1.2 Table Creation In Visual Data Manager

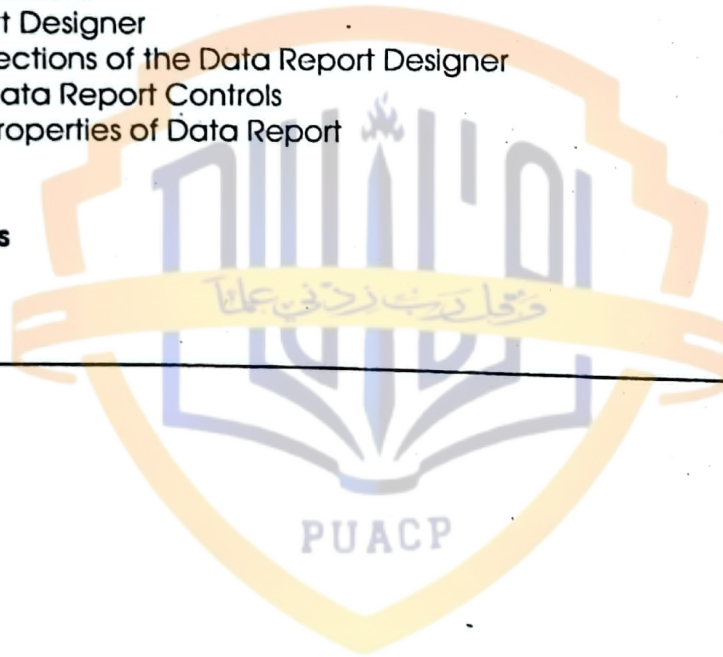
17.1.3 Creating Index In Visual Data Manager

17.2 Data Environment**17.3 Data Report Designer**

17.3.1 Sections of the Data Report Designer

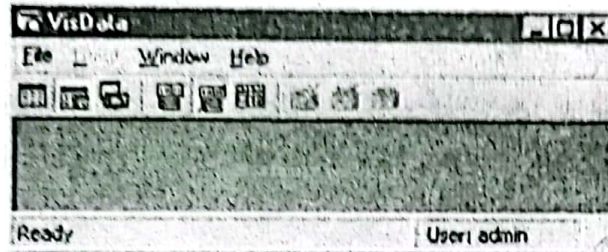
17.3.2 Data Report Controls

17.3.3 Properties of Data Report

Short Questions**Multiple Choices****Fill in the Blanks****True/False**

17.1 Visual Data Manager

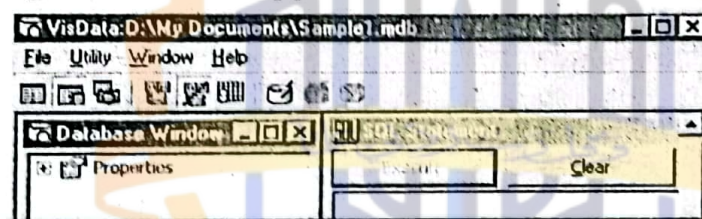
Visual Data Manager is a built-in tool in Visual Basic for creating databases. It is used to develop a database application without using MS Access. It can be started Visual Data Manager by clicking Visual Data Manager... from Add-Ins menu. The main IDE will appear as follows:



17.1.1 Creating Database in Visual Data Manager

The steps to create any database in Visual Data Manager are as follows:

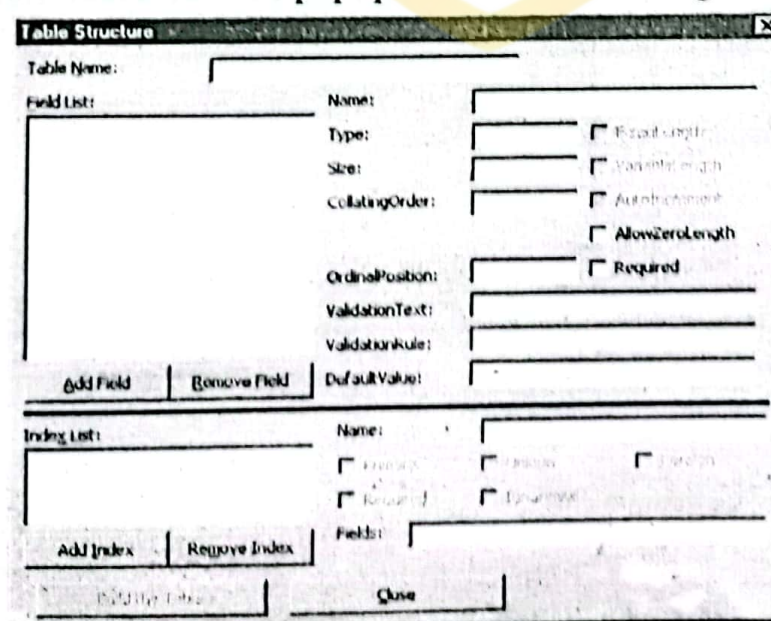
1. Start Visual Data Manager.
2. Select File > New > Microsoft Access > Version 7.0 MDB... A dialog box will appear.
3. Enter a name of database and click Save button. The database will be created and the following window will appear.



17.1.2 Table Creation in Visual Data Manager

You can create tables in a database by following these steps:

1. Right click in Database Window. A popup menu will appear.
2. Select New Table from the popup menu. The following window will appear.



Option	Description
Table Name	Name of the table to be created.
Field List	List of all fields in the table.

Table: Table Structure options

3. Enter the name of table and then click on Add Field button. The following window will appear.

Option	Description
Name	Name of the field.
Type	Data type of the field.
Size	Size of the field in bytes.
Required	Specify that the field cannot be blank.
Validation Text	The message to be displayed if the user enters invalid data.
Validation Rule	Validation condition for the field.
Default Value	Initial value of the field

Table: Options of Add Field window

17.1.3 Creating Index in Visual Data Manager

You can create index in tables by following these steps:

4. In Table Structure window, click on Add Index button. The following window will appear.

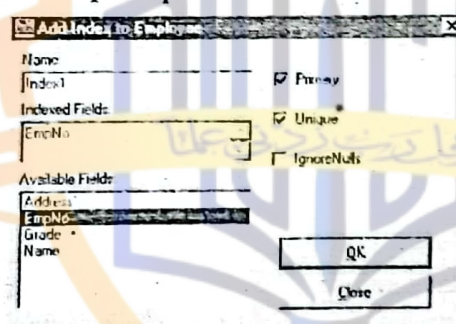
Option	Description
Name	Name of the index.
Indexed Fields	The fields to be indexed.
Available Fields	List of all fields in the table.
Primary	Specify whether the field is used as primary key or not.
Unique	Specify that the values in the field will be unique.
Ignore Null	Specify whether the field can contain Null value or not.

Practical 17.1

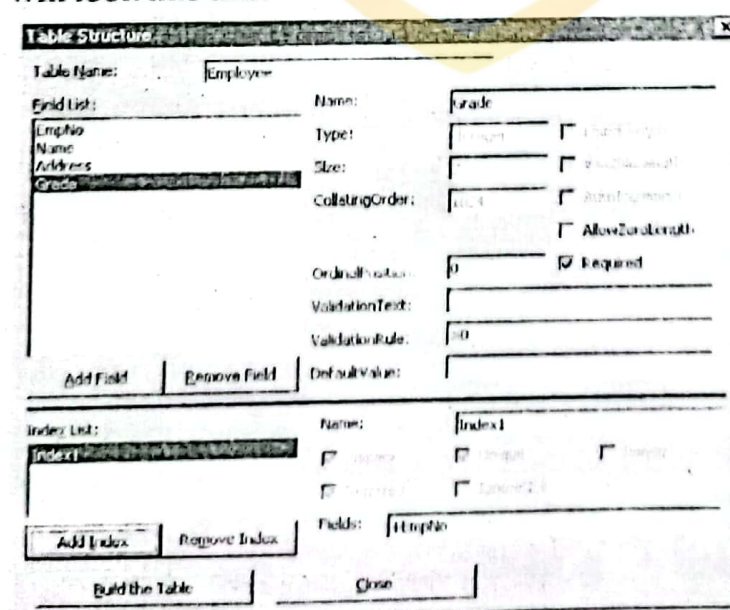
1. Start Visual Data Manager.
2. Select File > New > Microsoft Access > Version 7.0 MDB... A dialog box will appear.
3. Enter any name and click Save.
4. Right click in the Database Window and select New Table from the popup menu. The Table Structure box will appear.
5. Enter "Employee" in Table Name.
6. Click on Add Field button.
7. Create the following fields by following the steps discussed earlier.

Name	Type	Required	Validation Rule
EmpNo	Integer	Yes	>0
Name	Text	Yes	
Address	Text	Yes	
Grade	Integer	Yes	>0

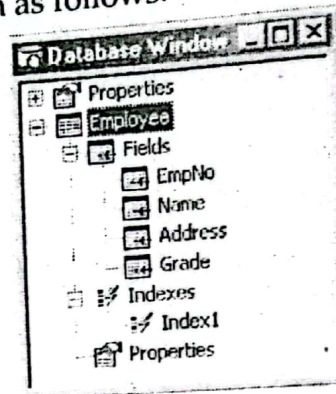
8. Close the window.
9. Click on Add Index button. A new window will appear.
10. Click on "EmpNo" in Available Fields box.
11. Enter "Index1" in Name field.
12. Select Primary and Unique options as follows:



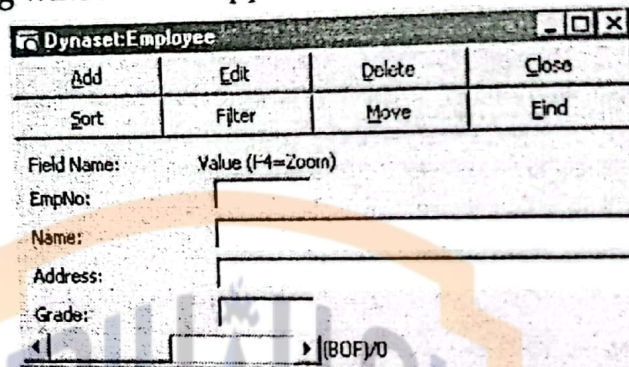
13. Click OK and then Close button. Index will be created and Table Structure window will look like this:



14. Click on **Build the Table** button. The table will be created and the Database Window will be shown as follows:

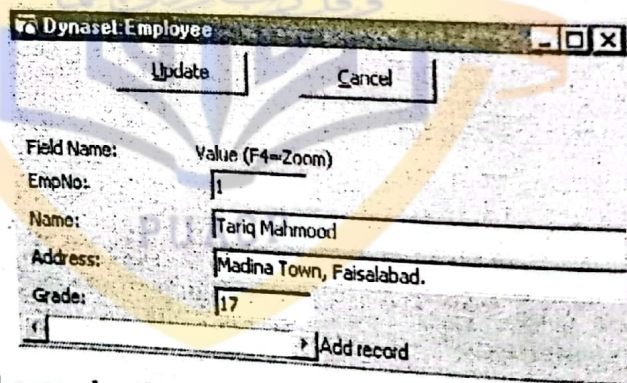


15. Right click **Employee** table and select **Open** or Double click on **Employee** table. The following window will appear.

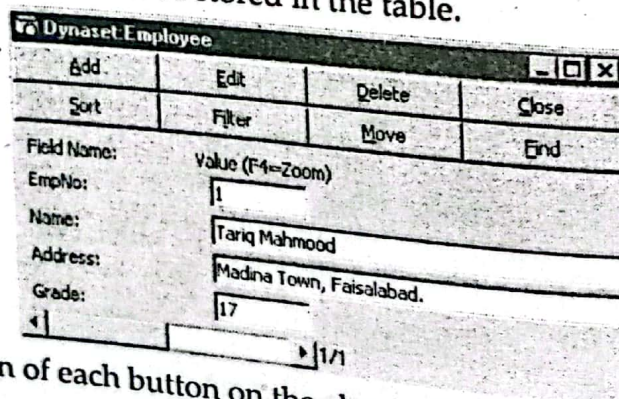


16. Click **Add** button. The new window will appear.

17. Add values in the fields and click **Update** button.



18. The entered record will be stored in the table.



A brief description of each button on the above window is as follows:

Add Button
This button is used to add new records in the table.

Edit Button

This button is used to edit the current record.

Delete Button

This button is used to delete the current record.

Close Button

This button is used to close the window.

Sort Button

This button is used to sort the records in the table. When you click this button, an input box appears. You can enter the name of the column according to which you want to sort the table. For example, if the user enters Name in the box and click OK, it will sort the table by Name field as follows:

Add	Edit	Delete	Close
Sort	Filter	Move	Find
Field Name:	Value (F4=Zoom)		
EmpNo:	1		
Name:	Ahsan Raza Sattar		
Address:	Peoples Colony, Fsd.		
Grade:	17		
1/4			

Filter Button

This button is used to filter records. When you click this button, an input box appears. Enter any condition to filter the table like:

Enter Filter Expression:

OK Cancel

The above condition will display one those records where EmpNo is greater than 1 as follows:

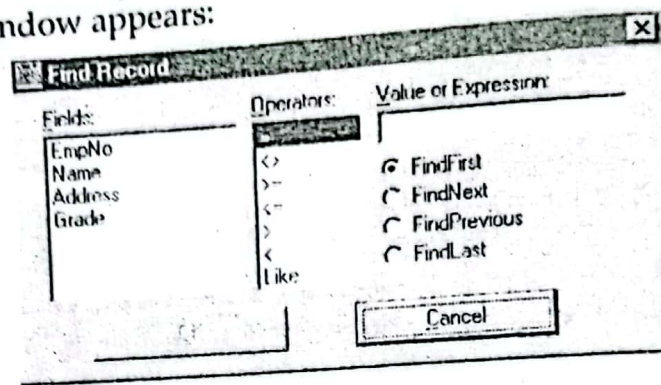
Add	Edit	Delete	Close
Sort	Filter	Move	Find
Field Name:	Value (F4=Zoom)		
EmpNo:	2		
Name:	Imran Samed		
Address:	Gulistan Colony, Faisalabad.		
Grade:	17		
1/3			

Move Button

This button is used to move forward or backward in the table. When you click this button, an input box appears. You can enter a positive number to move forward or negative number to move backward. For example if you enter -2, the control will move two records backward.

Find Button

This button is used to search a particular record in the table. When you click this button, the following window appears:



You can give any condition for searching the desired record. For example,

- Select EmpNo from Fields box.
- Select > from Operators box.
- Enter 2 in Value or Expression box.
- Click OK button. It will display the records whose EmpNo is greater than 2.

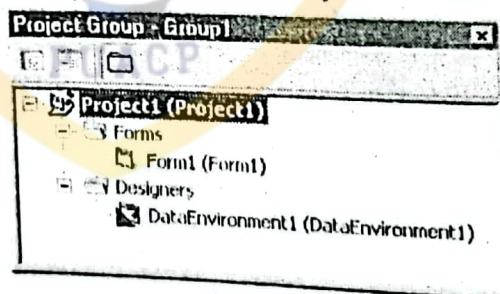
17.2 Data Environment

The Data Environment holds information required to access data from the databases. It contains one or more data connections. Each data connection refers to one or more data commands that represent a method for querying or modifying the database.

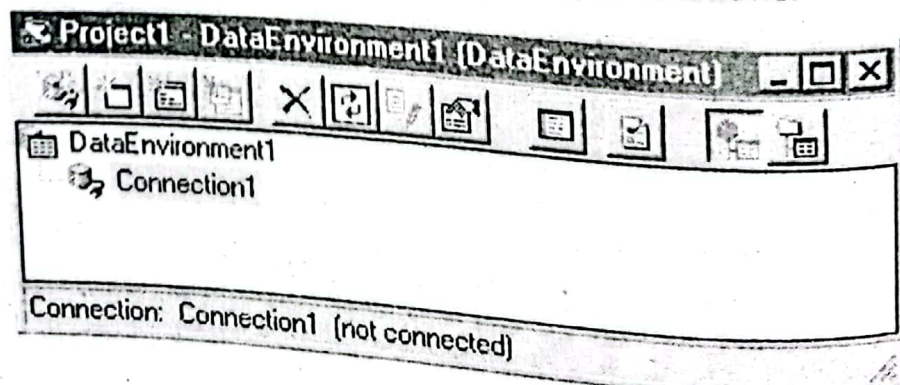
The Data Environment Designer provides a graphical way to connect to the Data Source. It is used to create and modify data environment of forms and reports of a project.

Practical 17.2

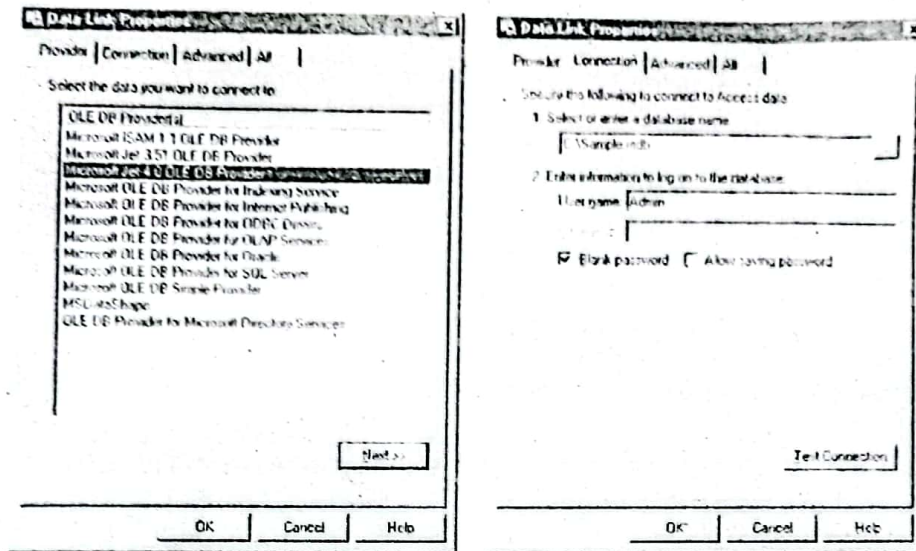
1. Start a new Standard EXE project.
2. Select Add Data Environment from Project. It will appear in project explorer.



3. The main window of Data Environment will be as follows:



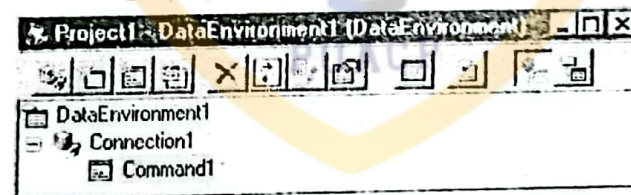
4. Right click on Connection1 and select Properties from popup menu. The Data Link Properties will appear.
5. Select Microsoft Jet 4.0 OLE DB Provider and click Next button.



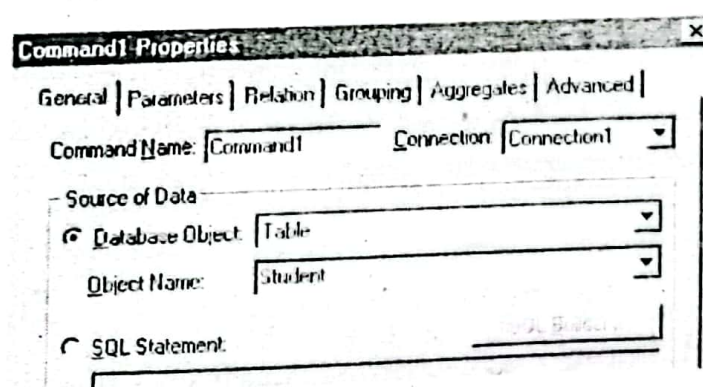
6. Select a database and click Test Connection button. The following message will appear.



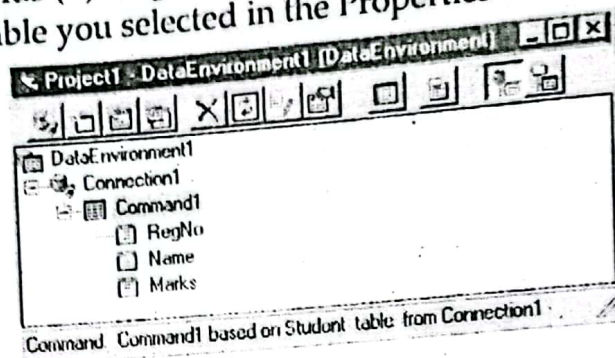
7. Click OK to close the message box.
8. Click OK to close the Data Link Properties dialog box.
9. Right click **Connection1** in Data Environment window and select **Add Command** from the popup menu. A command object will be added.



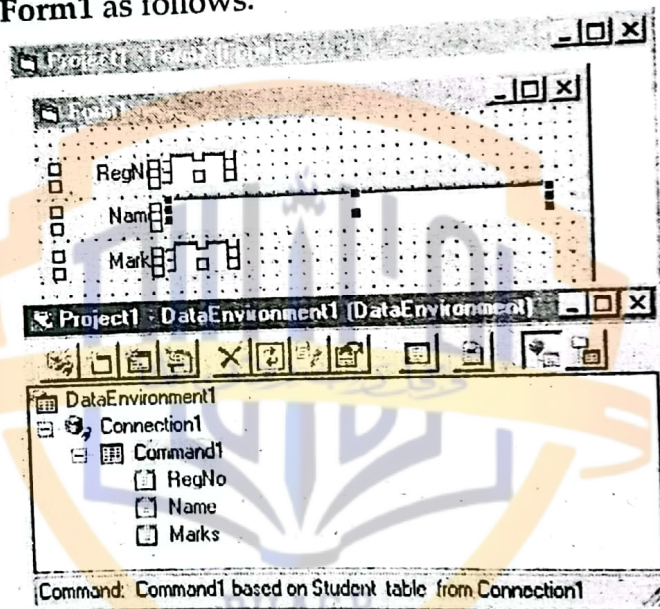
10. Right click on **Command1** and select Properties from popup menu. Properties window will appear:



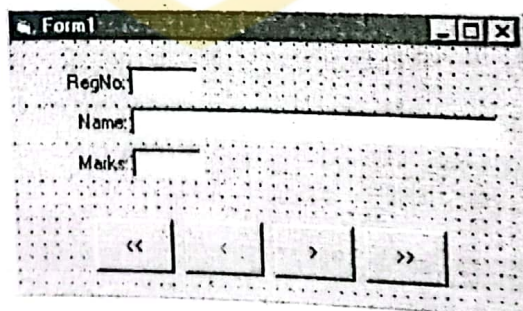
11. Click on **General** tab and select **Table** from **Database Object** list box and any table from **Object Name** list box.
12. Click **OK** button to close the **Properties** window.
13. Click on the **Plus (+)** sign on the left side of **Command1** object. It will contain the fields of the table you selected in the **Properties** window.



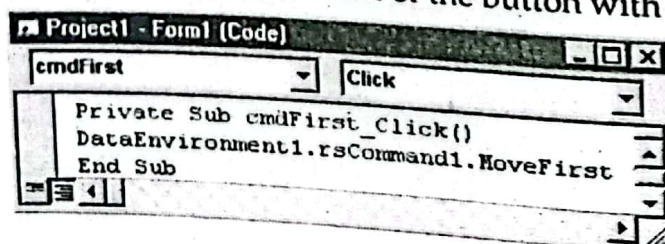
14. Arrange **Form1** and **Data Environment** in such a way that both should be visible simultaneously.
15. Drag the **Command1** object on the **Form1**. All fields in the **Command1** object will be added in **Form1** as follows:



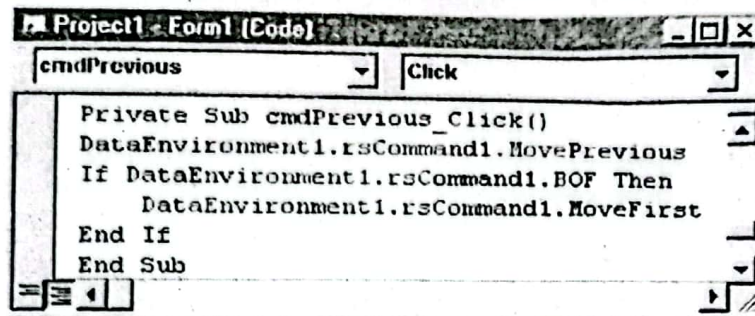
16. Place four button on the form as follows:



17. Add the following code in **Click** event of the button with caption "<<".



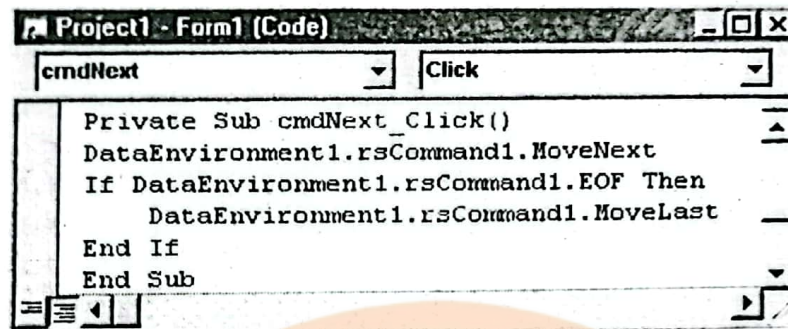
18. Add the following code in Click event of the button with caption "<".



```

Project1 - Form1 (Code)
cmdPrevious Click
Private Sub cmdPrevious_Click()
    DataEnvironment1.rsCommand1.MovePrevious
    If DataEnvironment1.rsCommand1.BOF Then
        DataEnvironment1.rsCommand1.MoveFirst
    End If
End Sub
  
```

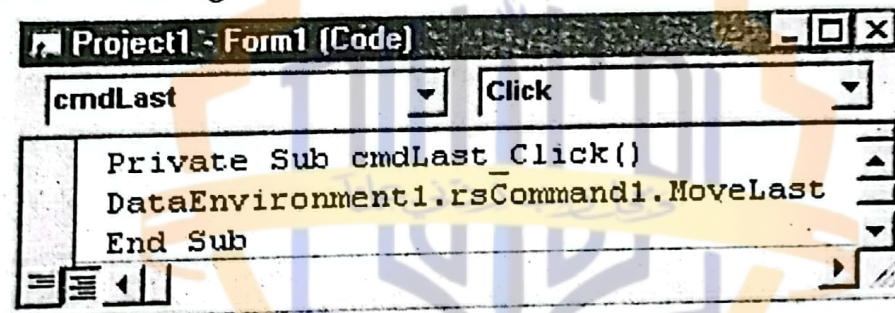
19. Add the following code in Click event of the button with caption ">".



```

Project1 - Form1 (Code)
cmdNext Click
Private Sub cmdNext_Click()
    DataEnvironment1.rsCommand1.MoveNext
    If DataEnvironment1.rsCommand1.EOF Then
        DataEnvironment1.rsCommand1.MoveLast
    End If
End Sub
  
```

20. Add the following code in Click event of the button with caption ">>".



```

Project1 - Form1 (Code)
cmdLast Click
Private Sub cmdLast_Click()
    DataEnvironment1.rsCommand1.MoveLast
End Sub
  
```

21. Run the project. You can navigate through the database by using the buttons.

17.3 Data Report Designer

The Data Report Designer consists of the following objects:

DataReport Object

The DataReport object consists of a visual designer and a code module. Visual designer is used to create the layout of a report. You can also add code to the designer's code module to programmatically format controls or sections contained by the designer.

Section Object

Each section of the Data Report designer is represented by a Section object. At design time, each section is represented by a header and section's pane where you can place controls.

Data Report Controls

Special controls that only work on Data Report designer are included with it. Visual Basic's intrinsic controls or any ActiveX controls cannot be used on Data Report designer. These controls are found in Visual Basic Toolbox but are placed on separate tab "DataReport".

17.3.1 Sections of the Data Report Designer

The default Data Report designer contains the following Sections:

Report Header

It is the text that appears at the very beginning of a report, such as the report title, author, or database name. If you want the Report Header to be the first page in the report, set its `ForcePageBreak` property to `rptPageBreakAfter`.

Page Header

It contains information that is displayed at the top of every page, such as report's title.

Group Header/Footer

It contains repeating section of data report. Each group header is matched with group footer. Header and footer pair is associated with a single **Command**.

Details

It contains the innermost "repeating" part (the records) of the report. The details section is associated with the lowest-level Command object in a Data Environment hierarchy.

Page Footer

It contains information that is displayed at the bottom of every page like page number.

Report Footer

It contains text that appears at the end of report such as summary information, address or contact name. The Report Footer appears between the last Page Header and Page Footer.

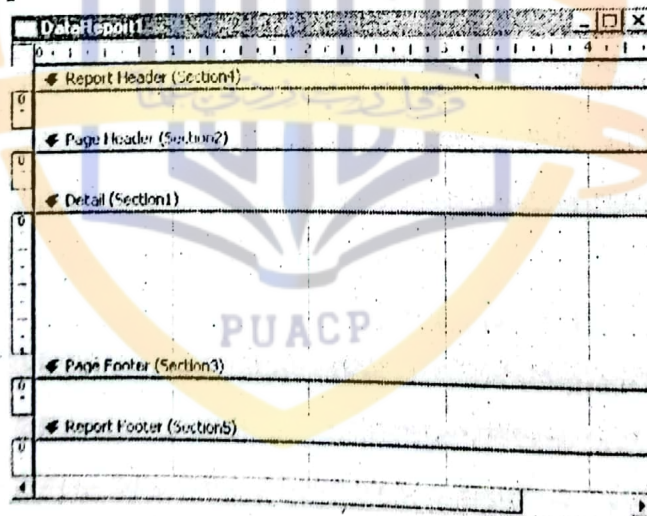


Figure: Data Report window

17.3.2 Data Report Controls

When a new Data Report designer is added to a project, the following controls are automatically placed in the Toolbox tab named DataReport:

- **TextBox Control** – It is used to format text or assign a DataFormat.
- **Label Control** – It is used to place labels on the report to identify fields or sections.
- **Image Control** – It is used to place graphics on your report. Note that this control cannot be bound to a data field.
- **Line Control** – It is used to draw lines on the report.
- **Shape Control** – It is used to place rectangles, triangles, or circles on a report.
- **Function Control** – It is a special text box to calculate values from different fields.

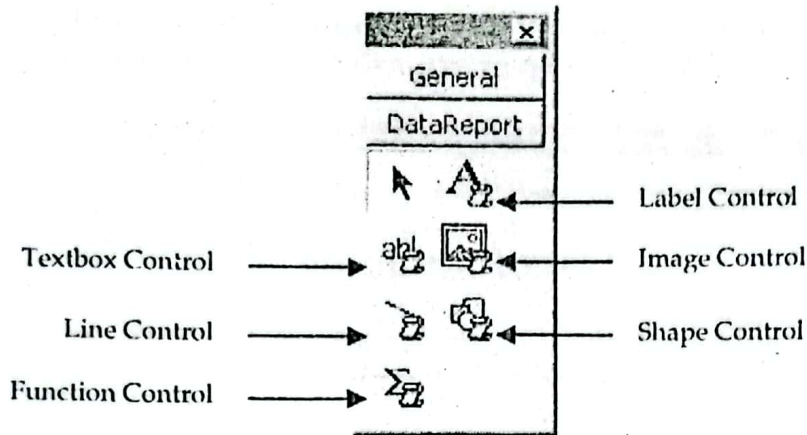


Figure: Data Report toolbox

17.3.3 Properties of Data Report

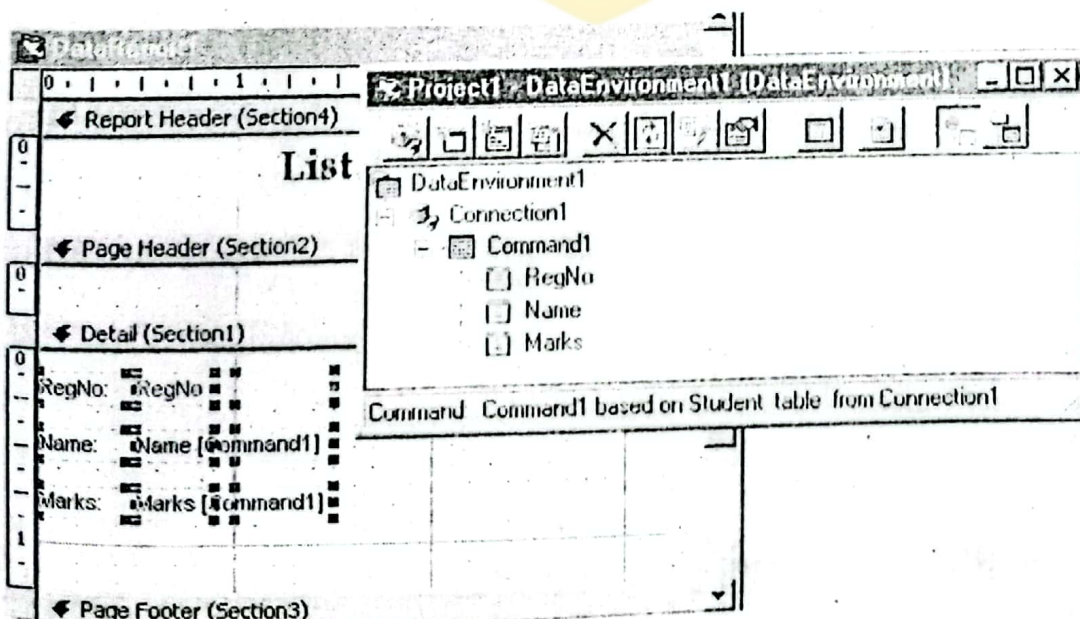
Some important properties of Data Report are as follows:

- **DataSource** – This property is used to specify the DataEnvironment object.
- **DataMember** – This property is used to specify the Command object in the specified DataEnvironment object.

Practical 17.3

Open the project of previous example and perform the following steps:

1. Select **Add Data Report** from Project menu.
2. Set **DataSource** property of DataReport1 to DataEnvironment1.
3. Set **DataMember** property of DataReport1 to Command1.
4. Place a label control in Report Header section and set its Caption to "List of Students".
5. Place a label control in Report Footer section and set its Caption to "Report generated by IT Series".
6. Arrange Data Environment window and Data Report window in such a way that both are visible simultaneously.
7. Drag Command1 object from Data Environment window in the Detail section of the Data Report window. It will create the following labels in the section.



8. The left most column in the section consists of labels used as headings of the table fields. Design the section as follows by dragging the labels with mouse.

DataReport1

Report Header (Section4)

List of Students

Page Header (Section2)

Detail (Section1)

RegNo	Name	Marks
RegNo	Name [Command1]	Marks [Command1]

Page Footer (Section3)

Report Footer (Section5)

Report generated by IT Series

9. Place a button on Form1 and add the following code in its Click event.

Project1 - Form1 (Code)

Command1 Click

```
Private Sub Command1_Click()
    DataReport1.Show
End Sub
```

10. Run the project and click the Report button on the form. The report will appear as follows:

DataReport1

Zoom 100%

List of Students

RegNo: 1	Name: Tariq Mahmood	Marks: 100
RegNo: 2	Name: Imran Saeed	Marks: 100
RegNo: 4	Name: Ahsan Raza	Marks: 100
RegNo: 5	Name: Usman Khalil	Marks: 90
RegNo: 3	Name: Tasleem	Marks: 90

Report generated by IT Series

Pages: 1

The above report displays field headings with each record. You can display the headings only once by moving the heading labels from Detail section to Page Header section as follows:

Report Header (Section4)		
List of Students		
Page Header (Section2)		
RegNo	Name	Marks
Detail (Section1)		
RegNo	Name [Command1]	Marks [Command1]
Page Footer (Section3)		
Report Footer (Section5)		
Report generated by IT Series		

Now run the project and click Report button. The report will appear as follows:

RegNo	Name	Marks
1	Tariq Mahmood	100
2	Imran Saeed	100
4	Ahsan F. Mustafa	100
5	Usman Khalil	90
3	Tasleem	90

Report generated by IT Series

Pages: 1

Short Questions

Q.1. What is the purpose of Visual Data Manager?

Visual Data Manager is a built-in tool in Visual Basic for creating databases. It is used to develop a database application without using MS Access.

Q.2. What is Data Environment?

The Data Environment holds information required to access data from the databases. It contains one or more data connections. Each data connection refers to one or more data commands that represent a method for querying or modifying the database.

Q.3. Is it possible to create a database using VB?

It is possible to create a database using VB. Visual Basic provides the facility of Visual Data Manager that can be used to create a database.

Multiple Choice

- A graphical way to connect to the Data Source is provided by:
 - Data Environment Designer
 - Data Report
 - Data Environment
 - None of the above
- The primary component of data environment is:
 - Data Report
 - Data Connection
 - Both of the above
 - None of the above
- The Report Header displays:
 - Text that appears at the very beginning of a report
 - Text that appears at the end of a report
 - Text that appears in the Footer of the report
 - None of the above
- A function control is placed in the:
 - Header Section
 - Footer Section
 - Group Footer Section
 - All of the above
- Which properties must be set to bind a control to a Data Environment?
 - DataSource and DataField
 - DataField and DataFormat
 - DataMember and DataField
 - DataSource, DataMember and DataField
- Which method of the DataReport enables the user to preview the report information?
 - PrintReport
 - PrintForm
 - Preview
 - Show
- A Data Project adds the following designers:
 - Data Environment
 - Data Report
 - Both a and b
 - Data Project
- The Data Report object represents:
 - Data Environment
 - Data Report designer
 - Both of the above
 - None of the above
- When a Data Report designer is added to the project, which tab is added to the Toolbox?
 - DataReport
 - DataEnvironment
 - Both a and b
 - None of the above

Answers

1. a	2. b	3. a	4. c
5. d	6. d	7. c	8. b
9. a			

Fill in the Blanks

- _____ is a built-in tool in Visual Basic for creating database.
- The _____ holds information required to access data from databases.
- The _____ provides a graphical way to connect to Data Source.
- The _____ is used to create database reports.
- Data Report Designer consists of _____ objects.

Answers

1. Visual Data Manager	2. Data Environment
3. Data Environment Designer	4. Data Report Designer
5. Three	

True / False

- Data Environment Designer is used to create and modify the data environment of forms and reports of a project.
- Data Report Designer consists of four objects.
- The data report object consists of a visual designer and a code module.
- Report Header contains information that is displayed at the bottom of every page.
- Report Header is the text that appears at the very beginning of a report such as report title, author or database name.
- Datasource is a property of Data Report that is used to specify the DataEnvironment object.

Answers

1. T	2. F	3. T	4. F
5. T	6. T		